Windows DSC

CYBEXER TECHNOLOGIES

What is DSC

- Improvement from security templates
  - Overall configuration
  - Usable also on non-domain joined computers
  - Can be used also for non windows devices

DSC requirements

- Windows Management Framework (WMF)
- WMF = PowerShell + WMI + WinRM
- At least WMF 4
- WMF 5 or later strongly suggested
- DSC related powershell commands
  - **Get-Command -Module PsDesiredStateConfiguration**

Create simple configuration

- Create DSC_test.ps1

```
Configuration DSCtest {
        Param { [String[]] $ComputerName = ”LocalHost”}
        Import-DscResource –ModuleName PSDesiredStateConfiguration
        Node $ComputerName {
                Registry RegDSCtest {
                        Ensure = ”Present”
                        Key = “HKEY_LOCAL_MACHINE\SOFTWARE\DSCkey”
                        ValueName = “TestingStuff”
                        ValueData = “0x0”
                        Hex = $True
                        ValueType = “Dword”
                        Force = “True”

                }
        }
}
```

Import the module

- Import-Module .\DSC_test.ps1
- Check from the functions folder if it is present
- dir function:\

Execute the new function

- DSCtest

Execute the new function

- DSCtest
- List the contents of current folder- notice a new folder ?

What is a MOF file

- Distributed Management Task Force (DMTF) introduced standard for Managed Object Format (MOF)
- Vendor neutral standard to define what the endresult HAS to be and not HOW it should be achieved
- Compiled MOF file is not a windows DSC explicit feature- MOF files can be created with anything you want- even texteditor but powershell is just more convenient

What are resource modules

- If MOF file describes what HAS to be achieved then a “translator” is needed to make it happen
- Resource modules are able to understand MOF desires and will do the heavy lifting neccessary to achieve it
- Resource modules in powershell are just powershell modules whose code is executed by a MOF file not a powershell script.

Apply DSC (MOF)

- Start-DscConfiguration –Path .\DSCtest –ComputerName localhost

DSC resource modules

- What kind of resources can you manage with DSC ?
  - Get-DscResource
- How to get more information about a resource ?
  - Get-DscResource –Name file -Syntax
- You dont have to write yourself- most probably the resource is already written by someone else !
- Find them online
  - Find-DscResource
- Find a more specific one
  - Find-DscResource |Where { $_.Name –Like ”*dns*”}

How to test the MOF

- Test the last applied DSC
  - Test-DscConfiguration -Detailed
- Compare against a chosen MOF
  - Test-DscConfiguration –ReferenceConfiguration .\path\to\Computer.mof
- Get information about DSC history
  - Get-DscConfigurationStatus -All

Restore or remove a MOF

- C:\windows\system32\Configuration contains multiple files:
  - Current.mof (current desired state- most recently applied configuration)
  - Previous.mof (previously applied configuration)
  - Pending.mof (about to be applied pending mof)
  - MetaConfig.mof (modified configuration settings for the Local Configuration Manager- might not exist !)
- Restore-DscConfiguration (restore previous MOF)
- Remove-DscConfigurationDocument –Verbose –Stage Current (can also be Pending or Previous according to Current,Previous,Pending MOF)

Who applies the MOF ?

- Local Configuration Manager (LCM) is responsible for applying the MOF files
- Get-DscLocalConfigurationManager
  - -ConfigurationMode
    - ApplyOnly
    - ApplyAndMonitor (Default)
    - ApplyAndAutoCorrect
  - -ConfigurationModeFrequencyMins 15..44640
  - -RebootNodeIfNeeded
  - -RefreshMode

Check MOF status locally/remotely

- Locally
  - Get-DscLocalConfigurationMonitor
- Remotely
  - Establish a CIM session first !
  - $session = New-CimSession –ComputerName computername
    Get-DscConfiguration –CimSession $session
    Remove-CimSession –CimSession $CimSess

Change LCM settings

- The following powershell will produce a META.MOF file to be applied

```
[DSCLocalConfigurationManager()] Configuration MetaConfig {
	Param ([String[]] $Computername = “Localhost”)
	Node $Computername {
		Settings {
			ConfigurationModeFrequencyMins = “15”
			ConfigurationMode = “ApplyAndAutoCorrect”
		}
	}
}
```